MIL-B-18797F
7 November 1983
SUPERSEDING
MIL-B-18797E
10 December, 1976

MILITARY SPECIFICATION

BURNERS, OIL, MECHANICAL-DRAFT, AUTOMATIC (UNDER 400,000 BTU/HR INPUT CAPACITY)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers automatic mechanical-draft oil burners intended primarily for firing low-pressure heating boilers, central warm-air furnaces, and water heaters.

1.2 Classification. Burners shall be of the following types and classes, as specified (see 6.2):

Type I - High-pressure, pressure atomizing, gun design.
Type II - Low-pressure, air atomizing, gun design.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-B-3180 - Boilers and Related Equipment, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4530

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STANDARDS

MILITARY

MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MS49007 - Nozzles, Oil Burner Pressure Atomizing.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

DC3 - Automatic Temperature Controls/Room Thermostats.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

UNDERWRITERS' LABORATORIES, INC. (UL)

UL 296 - Oil Burners.
UL 343 - Pumps for Oil-burning Appliances.
UL 372 - Primary Safety Controls for Gasand Oil-Fired Appliances.
UL 506 - Standard for Speciality Transformers.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 396 - Fuel Oils, Specification For.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The automatic burner shall consist essentially of a mechanical-draft fan or blower, ignition system, automatic primary-safety controls, and a fuel atomizing system. Either Type I or Type II burners will be acceptable unless a specific type is specified in the contract or order (see 6.2).

3.2 First article. When specified (see 6.2), the contractor shall furnish one complete burner for first article inspection and approval (see 4.2.1 and 6.3).

3.3 Standard commercial product. The burner shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the burner being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Design. Unless otherwise specified (see 6.2), burners shall be designed for firing grade 2 fuel oil of ASTM D 396. Burner design shall incorporate a positive displacement shaft driven fuel oil pump (see 3.9.1), an integral oil valve (see 3.9.3), automatic primary safety controls (see 3.9.5), a direct spark ignition system (see 3.9.8), and a shaft driven mechanical-draft fan or blower (see 3.9.9). Burners shall be designed for input capacity in British thermal units per hour (Btu/hr) or gallons per hour (gph) as specified in the contract (see 6.2).

3.5.1 Type 1. Type I burners shall be of the high-pressure gun design with an oil supply pipe and nozzle centrally located in the burner air tube. The burner shall be equipped with an oil pump, capable of raising the oil pressure to not less than 100 pounds-force per square inch gage (psig) to effect pressure atomization as the oil discharges from the nozzle into the combustion chamber (see 6.2).

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3.5.2 Type II. Type II burners shall be of the low-pressure design with an air/oil supply pipe and nozzle centrally located in the burner air tube. The burner shall be equipped with a low-pressure oil pump, (5 to 45 psig), and the required connections and fittings for connection to an external air supply to effect atomization as the air/oil mixture is discharged through the nozzle into the combustion chamber. Type II burners shall be capable of firing properly when connected to an external air supply of the characterictics specified, (see 6.2).

3.5.3 Design limitations When specified (see 6.2) burners shall be suitable for use in the equipment described in the contract. When the burner is to be installed within a combustion chamber that imposes a limitation on the flame size, flame shape, or the length of the flame tube, suitable drawings or descriptions will be included in the contract and the burner will be designed accordingly (see 6.2).

3.6 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.7 Performance. Burners shall meet the performance requirements of UL 296.

3.8 Compliance. Prior to approval of the first shipment, the contractor shall submit to the contracting officer, or his authorized representative, satisfactory evidence that the burner he proposes to furnish under this specification meets the requirements of UL 296.

3.8.1 Certification. Acceptable evidence of meeting the requirements of UL 296 shall be either the UL certification symbol or label, or a certified test report (see 6.4) from a recognized independent testing laboratory indicating the burner has been tested and conforms to UL 296. Such evidence must be acceptable to the contracting officer.

3.9 Burner components. Burner components not covered herein shall conform to the applicable requirements of UL 296.

3.9.1 Fuel oil pump. The fuel oil pump for burners shall be driven by either the blower shaft or the burner motor, shall be of the positive displacement type and be in accordance with UL 343, the pump shall have sufficient capacity to provide the total volume of oil, at the required pressure to con- tinuously fire the burner at it's full rated capacity. Single stage pumps may be used where the suction lift is 10 feet (ft) or less. For applications where the suction lift is greater than 10 ft., but no more than 15 ft., a two stage pump shall be specified, (see 6.2). For applications where low lubricity fuels such as JP-5 will be used, hardened steel pump gears shall be specified, (see 6.2).

3.9.2 Nozzles. Nozzles and strainers for Type I and Type II burners shall conform to the requirements of MS49007. Rated capacities, spray patterns, and angles of spray shall be limited to the values specified therein. Nozzle tips shall be corrosion resistant steel or shall be brass with a corrosion resistant steel insert for the orifice portion. Unless otherwise specified (see 6.2), burners shall have nozzle spray pattern as recommended by either the heating appliance manufacturer or burner manufacturer as applicable.

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3.9.3 Oil valve. Each burner shall be equipped with an oil valve located between the pump and the nozzle. Burners 240,000 Btu/hr and over may be equipped with an oil delay valve that is energized between 2 and 7 seconds after the burner is energized and shall close instantaneously when the burner motor is de-energized.

3.9.4 Thermostat. When specified (see 6.2), each burner shall be furnished with a room thermostat suitable for use with the primary control supplied with the burner. Thermostats shall conform to the applicable requirements of NEMA DC3.

3.9.5 Primary control. The primary safety control shall respond directly to flame properties, sensing the presence or absence of flame. In the event that flame failure occurs on initial start or during the firing cycle, the primary control shall be designed to shut down the burner or may recycle once after a 60 second shutdown following which an attempt to initiate a normal starting program is made and shall then lock out and require a manual reset. The primary safety control programming as designated in UL 372 for main burner establishing period shall not exceed 45 seconds and the flame failure response time shall not exceed 4 seconds.

3.9.6 Mounting. Burners shall be equipped with an adjustable pedestal-type base or with a flange or bracket type mounting at the option of the manufacturer unless a particular mounting is specified (see 6.2).

3.9.7 Refractories. When specified (see 6.2), burners shall be provided with specially preformed refractories for hearths.

3.9.8 Ignition system. The direct spark ignition shall have a nominal voltage of at least 10,000 volts and shall be of the intermittent type as defined in UL 296. High voltage ignition transformers shall meet the requirements of UL 506.

3.9.9 Fan or Blower. The fan or blower shall be shaft driven by the burner motor and shall be capable of delivering 125% of the air required for complete combustion when the burner is firing at the maximum setting. The air volume shall be adjustable within the total firing range of the burner.

3.10 Electrical requirements. Unless otherwise specified (see 6.2), burners shall be designed for operation on 120 nominal voltage, 2 pole, 3 wire with ground, 60 Hertz single-phase alternating current. Controls for burners shall be prewired to the maximum practicable extent. Wiring practices shall be in accordance with NFPA 70.

3.11 Electromagnetic interference characteristics. When specified, (see 6.2), the equipment shall conform to the electromagnetic interference characteristics requirements and test limits for class 3 equipment as specified MIL-STD-461.

3.12 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

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3.13 Treatment and painting Unless otherwise specified (see 6.2), the burner shall be treated and painted in accordance with the manufacturer's standard practice. All surfaces of the burner other than corrosion-resisting steel shall be protected against corrosion and present a neat appearance.

3.14 Marking. Burners shall be marked in accordance with UL 296. When specified (see 6.2), burners shall be additionally marked in accordance with MIL-STD-130, as directed by the contracting officer or his authorized representative.

3.15 Workmanship. The quality of workmanship shall meet the standards prevalent among manufacturers who normally produce equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.1.2 Standards compliance. The contractor shall make available to the contracting officer or his authorized representative evidence of compliance with the applicable standard(s) cited in 3.8. The Government reserves the right to examine and test all burners to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1)
b. Quality conformance inspection (see 4.2.2)

4.2.1 First article inspection. The first article inspection shall be performed on one complete burner unit when a first article is required (see 3.2, 6.2 and 6.3). This inspection shall include the examination of 4.3, tests of 4.4, and, when specified, the preproduction pack inspection of 4.5 (see 4.5 and 6.2). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, tests of 4.4, and the packaging inspection of 4.5.

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4.3 Examination. Each burner shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. Tests shall conform to applicable standards cited in 3.8. When specified (see 6.2), certification as specified in 3.8 may be submitted in lieu of performing the required tests.

4.4.1 Measurement of electromagnetic radiation. To determine conformance to the electromagnetic interference characteristics of 3.11, electromagnetic radiation shall be measured in accordance with MIL-STD-462. When suppressed to conform to the electromagnetic interference requirements of 3.11, the manufacturer may upon approval of the contracting officer furnish a certification in lieu of the test (see 6.4), that the burners meet the requirements, together with a list of the suppression devices installed. The list shall be sufficiently detailed to allow visual determination that the devices are installed.

4.5 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-B-3180. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-B-3180 with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

6.1 Intended use. Burners covered by this specification are in the category of domestic oil burners for applications having a range of firing rates up to an approximate maximum of 3 gph. The burners are intended for use with low-pressure steam and hot-water boilers, central warm-air furnaces, floor-mounted space heaters, oil-fired water heaters, and similar equipment. Burners are arranged for mounting outside of the firebox, and for firing into the firebox.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When a specific burner type is required (see 1.2, 3.1, 3.5.1, and 3.5.2).
c. Issue of document required if different than as specified (see 2.1.1).
d. When a first article sample is required for inspection and approval (see 3.2, 4.2.1, and 6.3).
e. When burners shall be designed for firing other than grade 2 fuel oil (see 3.5).

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f. Input capacity required (see 3.5).
g. External air pressure, volume and quality for type II burners, (see 3.5.2).
h. When required, description of oil-fired equipment on which burners will be installed and details of combustion chamber and air tube (see 3.5.3).
i. When two stage fuel pump is required (see 3.9.1).
j. When hardened steel pump gears are required (see 3.9.1).
k. When nozzles shall be different (see 3.9.2).
l. When a room thermostat is required (see 3.9.4).
m. When either a pedestal mounting, flange mounting, or bracket mounting is specifically required (see 3.9.6).
n. When preformed refractory hearths are required for burners and description and dimensions of hearth (see 3.9.7).
o. When electrical requirements shall be other than specified (see 3.10).
p. When electromagnetic interference suppression is required, (see 3.11).
q. When treatment for fungus resistance is required (see 3.12).
r. When treatment and painting is different from standard commercial practice, (see 3.13).
s. When marking of burners in accordance with MIL-STD-130 is required (see 3.14).
t. When certification may be submitted in lieu of performing the required tests (see 4.4).
u. When a preproduction pack inspection is required (see 4.2.1). and the time frame required for submission (see 4.2.1 and 4.5).
v. Levels of preservation and level of packing required (see 5.1).

6.3 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one complete burner. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved DD Form 1664, Data Item Description (DID), and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirements title	Applicable DID No.	Option
4.4.1	Certification Data/Report	UDI-A-23264B	
3.8.1	Certification Data/Report	UDI-A-23264B	

(DIDs related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, Acquisition Management Systems and Data Requirements Control List. Copies of DIDs required by the

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contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.6 Supersession data. This document differs from the superseded specifications in the following classifications:

MIL-B-18797E	MIL-B-18797F
Design A - High pressure, air atomizing, gun type.	Type I - High pressure, pressure atomizing, gun design.
Design B - Low-pressure-atomizing, gun type.	Type II - Low pressure, air atomizing, gun design.
Class 1 - Integral burner.	Deleted
Class 2 - Conversion burner.	Deleted
Custodians: Army - ME Navy - YD Air Force - 99	Preparing Activity: Navy - YD Project No. 4530-0032
Review Activity: DLA - CS Air Force - 84 Army - MD	
User Activity: Army - CE Navy - CG, MS	

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